

Syed Usman Bukhari

Telephone: 07462660889 | Email: usmanbukhari541@gmail.com | Portfolio: usmanbukhari.co.uk | Github : github.com/UsmanBuk
Linkedin : www.linkedin.com/in/syed-usman-bukhari-aideveloper

Senior AI Engineer with production experience building enterprise *RAG systems*, *LLM* orchestration pipelines, and scalable cloud infrastructure. Delivered AI solutions serving 1.4 million residents at NHS South Yorkshire ICB with zero hallucination incidents. Combines deep Azure AI expertise (AI Search, Document Intelligence, OpenAI) with strong Full Stack & DevOps foundations including CI/CD automation, Kubernetes, and Terraform. Proven track record improving system performance, automating complex workflows, and shipping production AI at scale.

Key Skills / Tech Stack

LangChain | LlamaIndex | LangGraph | RAG Systems | Azure OpenAI | Azure AI Search | Azure AI Agent SDK | Document Intelligence | Vector Search | Embeddings | Prompt Engineering | MCP | NLP | Computer Vision | PyTorch | TensorFlow | Hugging Face | MLOps | Azure Functions | CosmosDB | AWS Lambda | SageMaker | Terraform | Kubernetes | Docker | Serverless | Python | FastAPI | Django | TypeScript | REST | GraphQL | Microservices | Event-Driven Architecture | ETL Pipelines | Apache Airflow | PySpark | Snowflake | PostgreSQL | MongoDB | Databricks | CI/CD | Git | Linux | TDD

Career Summary

April 2026 - Present

Ai Engineer - Hudson and Hayes

- Developing production **voice agents** on the **Azure Ai Foundry** stack using **GPT-4o realtime API** for low-latency speech to speech , with function calling for in-call tool use and action execution
 - Integrating **Azure Communcation Services** for telephony and **Azure Speech Services** for STT/TTS, with custom knowledge grounding to deliver domain-specific responses across inbound and outbound call flows.
 - Building **evaluation and observability pipelines** to monitor call quality,latency, and tool use accuracy across deployed agents, enabling iterative prompt and agent refinement from production conversation data.
-

Feb 2026 - April 2026

Ai / Full Stack Engineer - Coex (Client : Ardonagh)

- **Designed and shipped a KYC onboarding platform end-to-end** for Ardonagh Specialty — React/TypeScript frontend, Python FastAPI backend, Azure-hosted — replacing a manual analyst workflow and cutting time-to-onboard from multiple days/weeks to hours.
 - **Built a multi-agent system on Azure AI Foundry (GPT-4o)** that autonomously gathers and reconciles client data across Companies House, LexisNexis WorldCompliance, and internal PAS, producing structured risk summaries that analysts review rather than assemble.
 - **Engineered resilient API integrations** with Companies House (PSC/UBO lookups), LexisNexis (sanctions, PEP, adverse media screening), and internal databases — including caching, retry, and rate-limit handling — keeping end-to-end agent runs under 30 seconds per case.
 - **Productionised the end-to-end workflow with Azure Logic Apps** as the orchestration backbone — monitoring inboxes, triggering on inbound client emails, routing attachments through Azure Document Intelligence to extract PDF and structured data, and automatically creating cases in MongoDB ready for agent enrichment — alongside the full-stack foundations (auth, audit logging, role-based access) needed to run agents safely in a regulated compliance environment.
-

May 2025 - Feb 2026

Gen AI Engineer & Data Architecture Specialist - NHS - South Yorkshire ICB

Greenfield Project

Production RAG System & LLM Orchestration

- Engineered production-grade **RAG system** using **Azure AI Agent SDK** with **LangGraph state-based orchestration**, LangChain custom retrieval chains and LlamaIndex for input sanitization/enhancement, implementing automated critique loops and **multi-source grounding (Bing + proprietary data)**. Achieved 90% accuracy validation against NHS clinical guidelines with zero hallucination incidents in production
- **Built production Azure AI Agent with real-time web grounding via Bing Search API**, enabling dynamic knowledge retrieval beyond static datasets. Implemented function calling patterns and tool orchestration to blend real-time web data with proprietary NHS knowledge base, reducing outdated information issues by 95% and enabling responses to emerging health

guidance within 24 hours of publication

- Integrated **Azure Document Intelligence** to automatically extract structured content from NHS policy PDFs, clinical guidelines, and governance documents—leveraging **OCR** to process scanned documents, parsing tables, key-value pairs, and unstructured text into searchable Azure AI Search indexes with custom field mappings, enabling policy-grounded responses with source citation capabilities.
- Implemented advanced vector search capabilities using **Azure AI Search** with 768-dimensional embeddings (OpenAI Ada-003), to build **hybrid search architecture** (lexical + semantic + real-time web) that improved service discovery accuracy by 85% and reduced query resolution from 12 minutes to under 30 seconds

Full-Stack AI Application | Microsoft Framework Enhancement

- Forked and extensively customized Microsoft's sample-app-aoai-chatGPT framework with **Azure AI Agent integration**, multi-modal search, context-aware dialogue management, **Bing grounding for real-time information**, and custom knowledge grounding and **Reflexion agent pattern** for self correcting retrieval.
- Deployed to production serving 3,000+ monthly active users with 94% query success rate and 4.8/5 satisfaction

MCP Server Integration | Patient-Facing AI Access

- Developed Model Context Protocol (**MCP**) server to expose RAG chatbot capabilities directly within the NHS South Yorkshire GP app, enabling patients to query local health services, commissioned care, and real-time guidance without leaving their existing care management interface

Azure MLOps & Data Infrastructure

- Designed event-driven data pipeline on **Azure Functions** with serverless microservices, queue triggers, and automatic failover, achieving **99.7% uptime** while processing 15,000+ daily records with sub-5-second latency
- Architected fault-tolerant web scraping infrastructure using **Crawl4AI** (Python) with custom middleware, extracting structured data from 50+ regional websites supporting 1.4M population across South Yorkshire

Technical Skills

- **Development:** Python, TypeScript, React, Quart(Async Flask), REST APIs, event-driven architecture
- **Cloud/DevOps:** Azure Functions, CosmosDB, Azure CI/CD, serverless architecture, microservices, Azure Durable Functions
- **Data:** Crawl4AI, ETL pipelines, vector databases, hybrid search, data standardization

June 2024 – April 2025 - Contractor

Sage Intacct full Stack developer & Solutions Architect – TSG/Dayta (Sage Partner)

- Designed **CI/CD pipelines (AWS CodeBuild, CodePipeline)** with integrated security practices—TDD, API authentication, and encryption standards—improving deployment reliability and compliance
- Architected serverless applications using **AWS Lambda** and Step Functions for event-driven workflow orchestration
- Built and deployed two-way **Sage Intacct** integration application on Docker/Kubernetes with automated CI/CD via GitHub and Jenkins
- Developed internal tools with **Flask** and **React** on AWS Amplify, automating client support workflows and improving developer productivity

Jan 2022 – Feb 2024-Contractor

Full Stack Engineer – Citadel Health

- Developed **RESTful APIs** using **Django REST Framework (DRF)** and **OpenAPI** to support high-volume LIMS workflows.
- Designed cloud-native solutions leveraging AWS services (**EC2, S3, RDS, Lambda, API Gateway, IAM**) to deliver robust and scalable LIMS applications.

Jan 2019 – Jan 2022

Software/Data Engineer – [Chalkys.com](https://chalkys.com) (Shopify)

- **Designed and orchestrated ETL pipelines using Apache Airflow and Databricks**, automating competitor price analysis & product data scraping workflows. Utilized **Databricks SQL** for real-time analytics and reporting dashboards, processing structured and semi-structured data with scheduled DAGs for reliable data processing. Implemented comprehensive monitoring using **Grafana** to track pipeline performance, with Databricks handling large-scale data transformations and aggregations.

Education and Professional Training

BSc Engineering Mathematics – University of Bristol

AWS Cloud Practitioner Certification & Solutions Architect Certification – eCareers

Data Engineering Program – Ai Core

Portfolio: usmanbukhari.co.uk : Live demos, open-source contributions, further contract roles and extended project history